Adichunchanagiri Institute of Technology
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Chikkamagaluru

MINI PROJECT SYNOPSIS PRESENTATION | BAI586 | 2026–2027

Development of an IoT and AI-Enabled Rover for Real-Time Crop Health Monitoring

PROJECT GUIDE

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TEAM

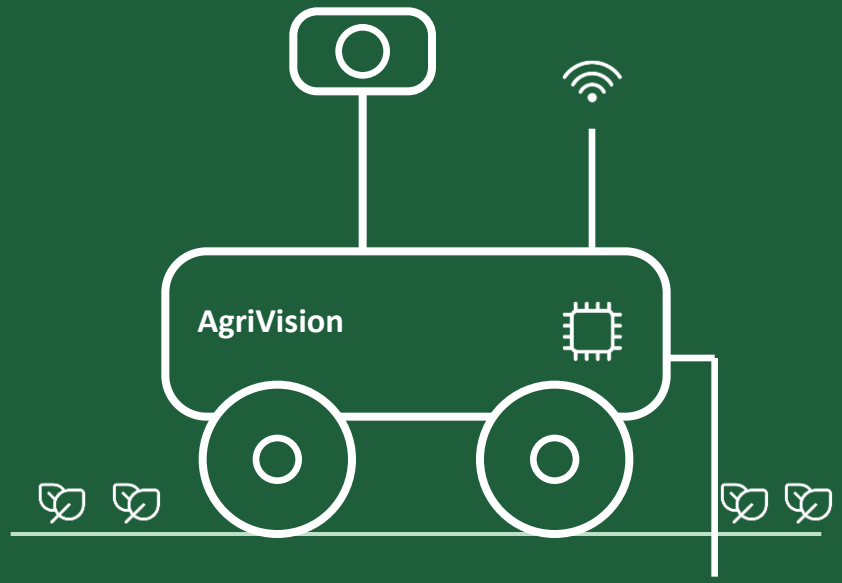
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Four-wheel monitoring rover (illustration)

AgriVision

IoT · AI · Robotics

Agenda

Five sections, from the field problem to the engineering plan

01	**Problem and research background**	Abstract · Introduction · Literature survey · Motivation · Problem statement
02	**Proposed system**	System overview · Objectives · Scope of the project
03	**Technical methodology**	Methodology · System architecture
04	**Implementation requirements**	Hardware components · Software and technologies
05	**Applications and future scope**	Applications and expected outcomes · Future scope · References

Abstract

From periodic manual inspection to continuous, mobile field monitoring

Existing systems for crop monitoring rely mainly on **periodic manual inspection** and on **fixed sensor nodes** that measure only a few points in a field. Problems that develop between inspections are noticed late, and sensor values alone do not show how the crop looks. Irrigation and crop-care decisions are therefore often made with **incomplete information**.

Proposed system is an IoT and AI-enabled rover that carries **environmental sensors and a camera** through the field. Two **ESP32 controllers** handle sensing and movement, an **ESP32-CAM** streams live video, and a Node.js backend stores the readings in **MongoDB** and presents them on a **React dashboard**. This working prototype is the foundation for **AI-assisted crop analysis** and decision support, the next stage of the project.

Sense
Sensors and a camera travel on the rover

● Built

Send
Wi-Fi telemetry and live video

● Built

Store and show
Backend, MongoDB and web dashboard

● Built

Analyse
AI-assisted crop analysis

○ Proposed

Introduction

Timely field information is the link between a problem and its remedy

Field conditions change

Crops face disease, pests, nutrient deficiency, water stress and weather extremes.

Detection is delayed

Periodic manual inspection is labour-intensive and hard to scale to large or remote fields.

Avoidable risk grows

Problems that develop between inspections lead to crop loss and wasted water.

Integrated monitoring is needed

Sensing, imaging and software must work together in the field.

Five complementary roles in the proposed system

Mobile robotics

Carries the sensors and camera to the crop

IoT sensing

Measures soil and weather conditions as the rover moves

Computer vision

Turns field images into usable observations

Environmental data

Adds rainfall and weather context

Farmer-facing software

Presents readings and alerts in a clear form

Literature Survey

Theme 1 – IoT sensing, edge computing and data-driven farming

Research theme	Authors and title	Proposed method	Drawback
IoT-based smart agriculture	**[1] Tzounis et al., 2017** *Internet of Things in agriculture, recent advances and future challenges* **[2] Elijah et al., 2018** *An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges*	Wireless sensor networks and IoT platforms monitor soil, climate and crop parameters, with remote access to the data.	Most deployments use fixed sensor nodes that cover limited points and give no visual crop information.
Edge and embedded intelligence	**[8] Shi et al., 2016** *Edge computing: Vision and challenges*	Data is processed close to its source to reduce latency and bandwidth.	Low-power microcontrollers limit on-device analysis, so a balanced edge–server design is needed.
Data-driven farming	**[9] Wolfert et al., 2017** *Big data in smart farming – A review* **[10] Kamilaris and Prenafeta-Boldú, 2018** *Deep learning in agriculture: A survey*	Big-data and deep-learning approaches combine weather, satellite and field data to support farm decisions.	Such data are rarely combined with ground-level rover observations in low-cost systems.

Takeaway: Sensing and data platforms are well studied, but they stay in fixed positions and carry no ground-level visual information.

Literature Survey

Theme 2 – Computer vision, irrigation sensing and agricultural robotics

Research theme	Authors and title	Proposed method	Drawback
AI and computer vision for crops	**[3] Mohanty, Hughes and Salathé, 2016** *Using deep learning for image-based plant disease detection* **[4] Ferentinos, 2018** *Deep learning models for plant disease detection and diagnosis*	Deep convolutional neural networks identify plant diseases from leaf images with strong reported results.	Many models are trained on controlled datasets and are seldom linked with live field sensing.
Sensor-based irrigation and soil monitoring	**[5] Ayaz et al., 2019** *Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk* **[6] Liakos et al., 2018** *Machine learning in agriculture: A review*	Soil-moisture and weather sensing are used to schedule irrigation and reduce water wastage.	Decisions often depend on a single parameter, without combining weather and crop condition.
Agricultural robotics	**[7] Bechar and Vigneault, 2016** *Agricultural robots for field operations: Concepts and components*	Field robots have been developed for inspection, spraying and harvesting.	Many platforms are costly and target a single operation rather than integrated monitoring.

Takeaway: Vision, irrigation sensing and robotics are usually studied separately, rarely as one field-monitoring system.

Research Gap and Project Positioning

The gap is integration: sensing, imaging, mobility and software are usually studied separately

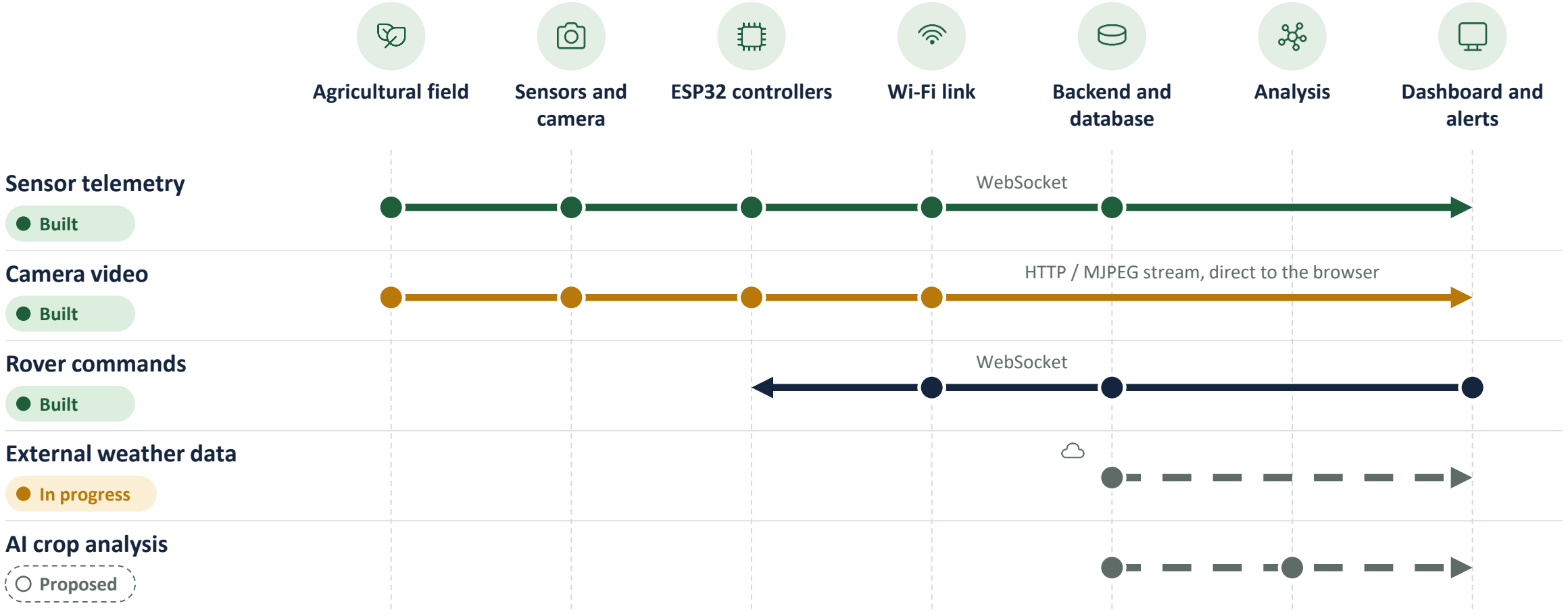
EXISTING APPROACH	LIMITATION	HOW AGRIVISION ADDRESSES IT	
Fixed IoT sensor nodes References [1], [2]	Cover limited points and give no visual crop information	Sensors and a camera travel through the field on a mobile rover	Built
Image-based disease models References [3], [4]	Trained on controlled datasets; seldom linked with live field sensing	Field images are captured together with sensor readings; model integration is the next stage	Proposed
Sensor-based irrigation References [5], [6]	Decisions often rest on a single parameter	Soil-moisture and rain readings are shown together; weather data is being added	In progress
Agricultural robots References [7]	Costly platforms built for a single operation	A low-cost ESP32 rover dedicated to field monitoring	Built
Edge and data-driven systems References [8]–[10]	Microcontrollers limit on-device analysis; ground-level data is rarely included	ESP32 nodes acquire the data; the backend stores and processes it centrally	Built

This is the integration gap identified in our literature review. It is not a claim that no existing system combines these capabilities.

Status: Built | In progress | Proposed

Overview of the Proposed System

Four data paths link the field to the farmer; AI analysis is the planned fifth



Arrows show the direction of travel. A dot marks each stage a path passes through.

Status: Built | In progress | Proposed

Motivation

Why a mobile, integrated monitoring platform is worth building

Field problems rarely wait for the next inspection.

Four reasons to bring sensing, imaging and software to the field together

1 Problems develop between manual inspections

They are noticed late, when they are harder to manage.

Frequent, repeatable passes by the rover

2 Sensor readings alone do not show how the crop looks

Symptoms on leaves and plants go unseen.

A live camera view alongside the readings

3 Watering needs soil and weather information together

Deciding on one value wastes water or stresses the crop.

Soil-moisture and rain readings in one place

4 Raw data is hard to act on

Numbers do not become decisions by themselves.

A dashboard that shows status and alerts clearly

Problem Statement

What must be solved, and what goes in and comes out of the system

Periodic manual inspection and **isolated field measurements** can delay awareness of crop and environmental problems. An **integrated field-monitoring platform** is needed to collect environmental readings and crop imagery, communicate the data reliably, and present timely field information through an accessible software interface.

INPUT

Environmental sensor readings	Built
Field imagery from the ESP32-CAM	Built
External weather data	In progress

PROCESS

Embedded acquisition on ESP32	Built
Wi-Fi and WebSocket communication	Built
Backend processing and MongoDB storage	Built
AI / computer-vision analysis	Proposed

OUTPUT

Live field status	Built
Environmental alerts	In progress
Crop and field insights	Proposed
Decision-support information	In progress

Objectives

Eight engineering objectives, each with a check that shows it is met

01 **To develop a four-wheel mobile rover platform for field monitoring**

Check: drives forward, in reverse and turns under remote command

02 **To integrate environmental and soil sensors with a dedicated Sensor ESP32**

Check: every sensor type reports readings

03 **To implement reliable Wi-Fi telemetry and rover control over WebSocket**

Check: readings and commands are exchanged in real time

04 **To integrate ESP32-CAM video for camera-based field observation**

Check: live video is visible in the browser

05 **To build backend APIs and persistent data storage in MongoDB**

Check: telemetry is stored and can be retrieved

06 **To develop a web dashboard for live monitoring and rover control**

Check: readings, video and controls appear on one screen

07 **To integrate AI and computer-vision analysis of field images as the next stage**

Check: the model is evaluated on representative field images

08 **To test and evaluate the integrated hardware and software prototype**

Check: an end-to-end test of the complete system

Scope of the Project

What the prototype covers today, and what is still being added

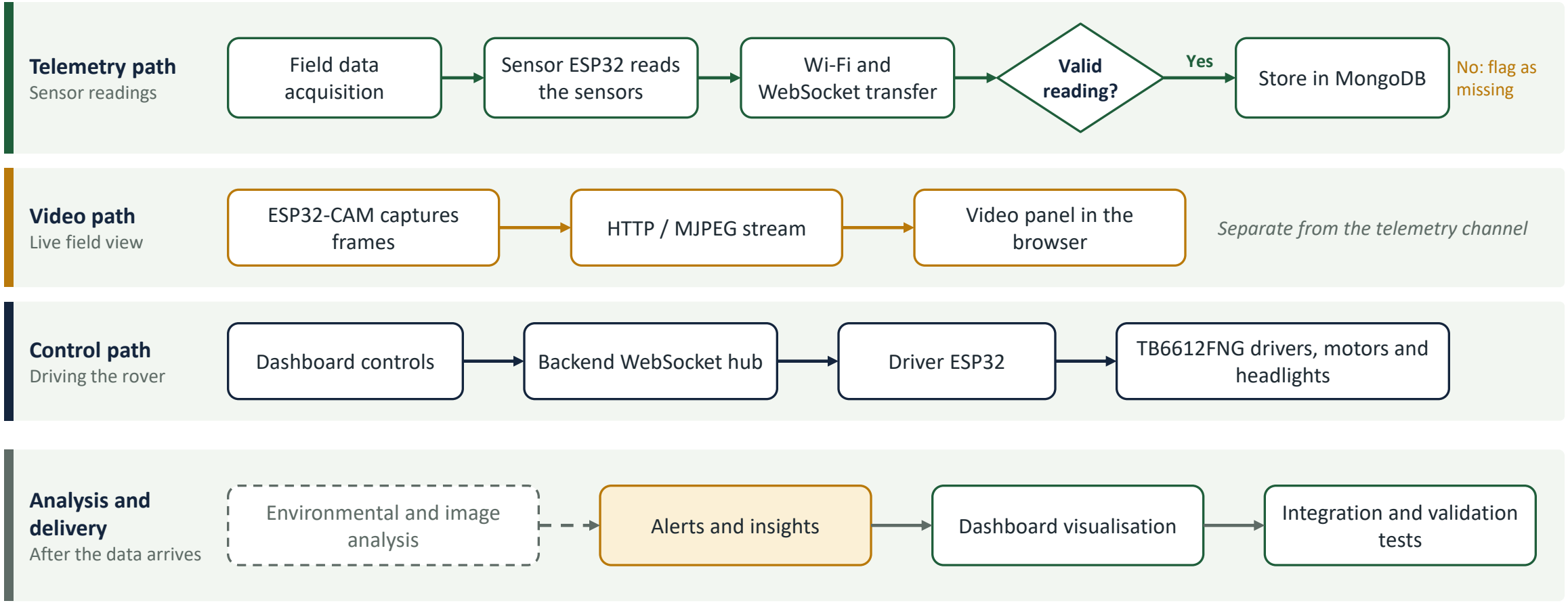
#	SCOPE AREA	WHAT IT COVERS	STATUS
1	**Mobile field monitoring**	Rover-based inspection of the field under remote control	Built
2	**Environmental and soil sensing**	Temperature, humidity, soil moisture, rain and flame	Built
3	**Camera-based field observation**	Live ESP32-CAM video of the crop and field	Built
4	**Rover movement and obstacle sensing**	Four-wheel drive; four ultrasonic distance sensors	Built
5	**Wireless communication and backend**	Wi-Fi, WebSocket telemetry and commands, Node.js services	Built
6	**Web monitoring and decision support**	Live dashboard and controls; alerts and guidance being added	Built, In progress
7	**AI and computer-vision analysis**	Crop-image analysis for disease, pest and stress indicators	Proposed
8	**Data storage and historical review**	MongoDB storage; trend review of past readings being added	Built, In progress

Outside this prototype: autonomous navigation, GPS field mapping and automatic irrigation. These are listed under Future Scope.

Status: Built · In progress · Proposed

Methodology

Three paths run side by side, then meet at analysis and the dashboard



The three paths run at the same time. Telemetry and video do not wait for each other.

Status: Built | In progress | Proposed

Hardware Components

Three on-board nodes share the work: sensing, driving and seeing

Sensor ESP32

Reads every sensor and sends telemetry

DHT11 air temperature and humidity
Soil-moisture sensor soil water level
Rain sensor rainfall detection
Ultrasonic sensors ×4 obstacle distance
5-channel flame sensor fire detection
Buzzer audible alert

Mechanical and power

Carries and powers the modules

Four-wheel chassis mounts all modules
Battery pack on-board supply
Buck converter regulated voltage for controllers and sensors

Driver ESP32

Drives the rover and its lights

TB6612FNG drivers ×2 motor direction and speed
BO DC motors ×4 four-wheel drive
Headlights front and rear visibility

Camera subsystem

A separate node for live video

ESP32-CAM live field video over Wi-Fi
Servo mount camera positioning

Why three nodes?

Sensing and driving run on separate ESP32 boards, so motor control does not interrupt sensor readings. The ESP32-CAM streams video as its own node.



Labelled layout of the rover

Concept render from the SIH submission, not a photograph. To be replaced with a photo of the assembled prototype.

Software and Technologies

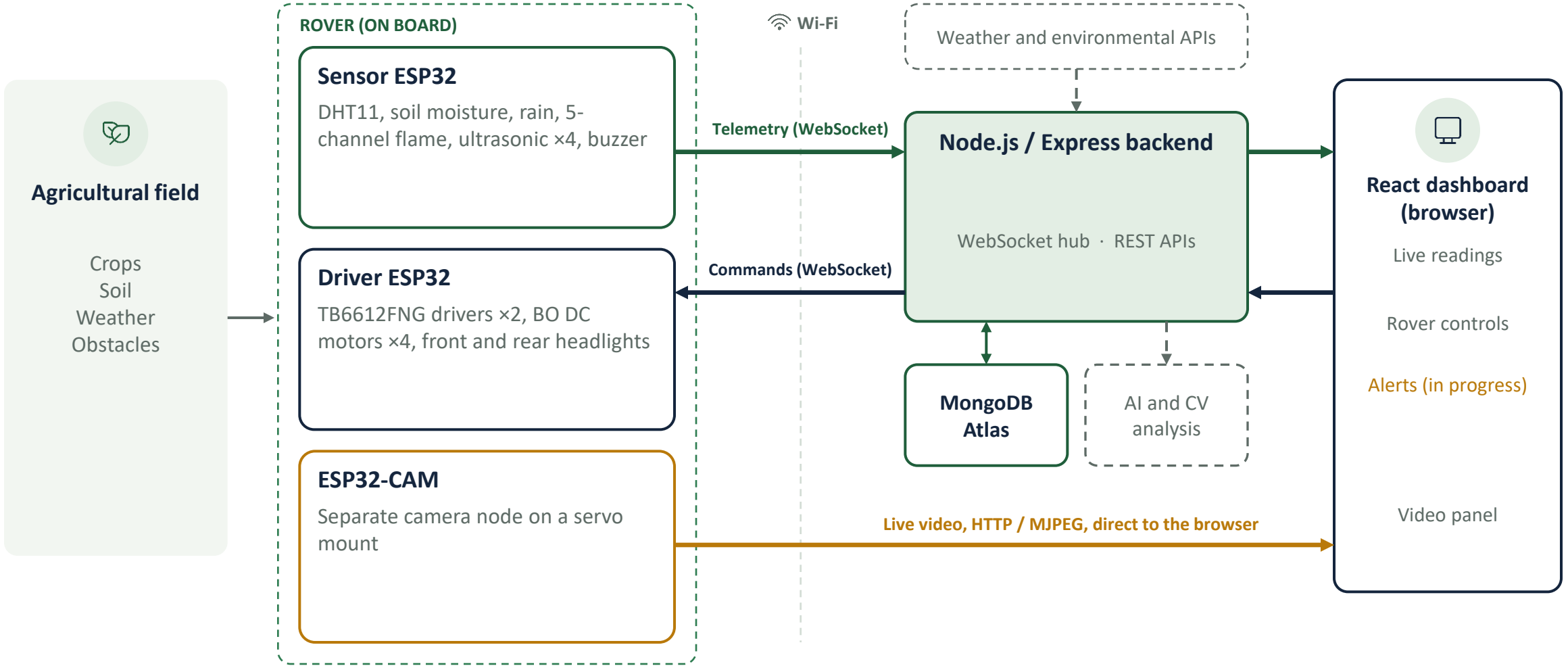
Each layer has one clear job; the status column shows what is running today

Frontend	React, Vite, Tailwind CSS	Dashboard with live readings, a video panel and rover controls	Built
Backend	Node.js, Express, REST APIs, WebSocket hub	Receives telemetry, relays rover commands and serves stored data	Built
Database	MongoDB Atlas	Persistent storage, reached only through the backend	Built
Embedded firmware	ESP32 (C / C++), WebSocket client	Sensor acquisition, motor control and headlight control	Built
Camera	ESP32-CAM, HTTP / MJPEG	Live video stream opened directly in the browser	Built
AI and computer vision	Python, OpenCV	Crop-image analysis; a model is still to be trained and evaluated	Proposed

No custom-trained model is deployed yet. Python and OpenCV are the planned tools for the analysis stage.

System Architecture

Three on-board nodes, one backend and one dashboard, joined by separate flows



Applications and Expected Outcomes

Where the rover is useful, and what the project will deliver

Where it can be used

Routine field inspection
Repeated passes instead of occasional walks

Remote observation
Field conditions viewed from the dashboard

Soil-moisture and rainfall awareness
Readings that support watering decisions

Fire and obstacle awareness
Flame and ultrasonic sensing while the rover runs

Camera-assisted observation
A live view of the crop and the field

Review of stored telemetry
Past readings kept in MongoDB

Expected outcomes (deliverables)

Integrated mobile rover prototype	Built
Sensor telemetry visible on the web dashboard	Built
Camera feed through the streaming path	Built
Centralised backend and MongoDB storage	Built
Environmental alerts and guidance	In progress
A foundation for AI-assisted crop analysis	Proposed

Outcomes are prototype deliverables. Effects on crop yield or water use will be stated only after field measurement.

Future Scope

A three-stage roadmap beyond the current prototype

1 — 2 — 3

STAGE 1

Reliability and field validation

- Reliable power for long runs
- Moisture protection for the electronics
- Sensor calibration
- Repeatable field tests
- Measured communication reliability

STAGE 2

Intelligence

- Validated crop-image models
- Disease and pest classification
- Location-aware environmental intelligence
- Seasonal crop-suitability advice
- Stronger water-stress analysis

STAGE 3

Advanced robotics

- GPS-based field mapping
- Autonomous path planning
- Better mobility on wet ground
- Closed-loop irrigation control, where the hardware permits
- More capable edge-computing hardware, if required

○ Future

Every item on this slide is a future enhancement. None is part of the current prototype.

References

IEEE format, as listed in the project synopsis

[1] A. Tzounis, N. Katsoulas, T. Bartzanas and C. Kittas, **"Internet of Things in agriculture, recent advances and future challenges,"** *Biosystems Engineering*, vol. 164, pp. 31–48, 2017.

[2] O. Elijah, T. A. Rahman, I. Orikumhi, C. Y. Leow and M. N. Hindia, **"An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges,"** *IEEE Internet of Things Journal*, vol. 5, no. 5, pp. 3758–3773, 2018.

[3] S. P. Mohanty, D. P. Hughes and M. Salathé, **"Using deep learning for image-based plant disease detection,"** *Frontiers in Plant Science*, vol. 7, Art. 1419, 2016.

[4] K. P. Ferentinos, **"Deep learning models for plant disease detection and diagnosis,"** *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.

[5] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour and E.-H. M. Aggoune, **"Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk,"** *IEEE Access*, vol. 7, pp. 129551–129583, 2019.

[6] K. G. Liakos, P. Busato, D. Moshou, S. Pearson and D. Bochtis, **"Machine learning in agriculture: A review,"** *Sensors*, vol. 18, no. 8, Art. 2674, 2018.

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[13] G. Bradski, **"The OpenCV Library,"** *Dr. Dobb's Journal of Software Tools*, 2000.

CONCLUSION

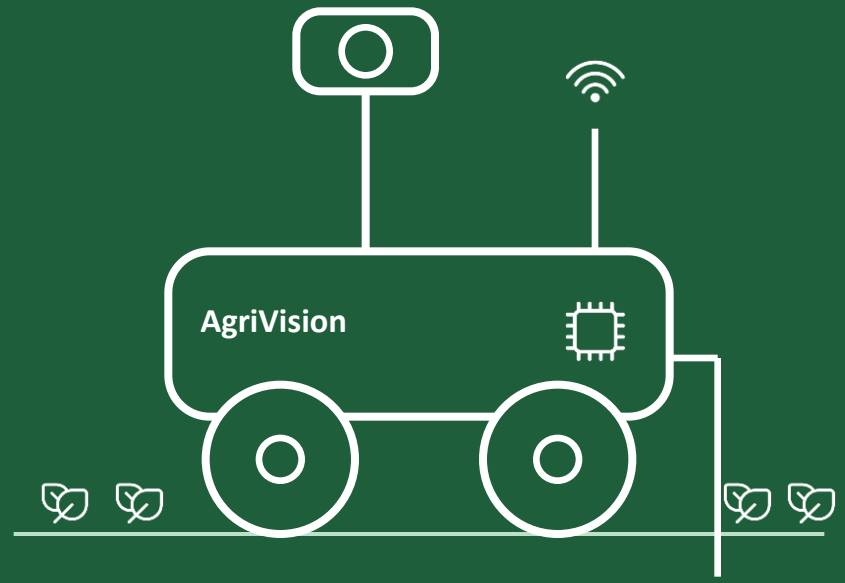
Connecting field sensing, mobile observation and software-based analysis to support timely agricultural monitoring.

Built	Rover, sensing, live video, dashboard and storage
In progress	Alerts, weather data and historical review
Proposed	AI-assisted crop analysis

Thank you

Questions and discussion

Adichunchanagiri Institute of Technology, Chikkamagaluru
Department of Artificial Intelligence and Machine Learning



Four-wheel monitoring rover (illustration)

AgriVision

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